INTRODUCTION TO UNIVERSITY MATHEMATICS

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Abstract. Notes for the University of Helsinki course MAT11001, taught in Finnish from 31.08.2026–17.10.2026.

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1. Course information

The official course information is available on the University of Helsinki course page.

2. Lecture notes

2.1. **31.08.2026.** The first lecture was mostly spent going over the course structure and other general admin topics. We did go over some preliminary definitions and examples, which will follow.

2.1.1. *Notational conventions.*

Definition 2.1. An integer $p > 1$ is *prime* if its only positive divisors are 1 and p.

Date: Autumn 2026.

Definition 2.2. The following basic sets of numbers will be used throughout:

$$\begin{aligned}
\mathbb{N} &= \{0, 1, 2, \ldots\} \\
\mathbb{Z} &= \{\ldots, -2, -1, 0, 1, 2, \ldots\} \\
\mathbb{Z}^+ &= \mathbb{N} \setminus \{0\} \\
\mathbb{Q} &= \left\{ \frac{p}{q} : p, q \in \mathbb{Z},\ q \neq 0 \right\} \\
\mathbb{R} &= \text{the set of real numbers} \\
\mathbb{C} &= \{a + bi : a, b \in \mathbb{R}\}, \quad i^2 = -1 \\
\mathbb{P} &= \{p \in \mathbb{Z} : p \text{ is prime}\}
\end{aligned}$$

2.1.2. *Principle of mathematical induction.* Here we state the fundamental principle of mathematical induction without proof, and use it freely thereafter.

Theorem 2.3 (Principle of mathematical induction)**.** *Let $n_0 \in \mathbb{N}$, and let $P(n)$ be a statement for each $n \in \mathbb{N}$ with $n \geq n_0$. Suppose that*

(1) $P(n_0)$ is true, and
(2) for every $n \geq n_0$, $P(n)$ implies $P(n+1)$.

Then $P(n)$ is true for every $n \geq n_0$.

Proof. Omitted. □

Example 2.4. We prove by induction that

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$$

for every integer $n \geq 1$.

For $n = 1$, the identity holds because

$$\sum_{k=1}^{1} k = 1 = \frac{1(1+1)}{2}.$$

Now suppose that the identity holds for some integer $n \geq 1$. Then

$$\begin{aligned}
\sum_{k=1}^{n+1} k &= \sum_{k=1}^{n} k + (n+1) \\
&= \frac{n(n+1)}{2} + (n+1) \\
&= \frac{(n+1)(n+2)}{2}.
\end{aligned}$$

Thus the identity also holds for $n+1$, and hence for every integer $n \geq 1$. □

2.2. **03.09.2026.**

2.2.1. *Divisibility of integers.* We start by defining the precise meaning of *even* and *odd* numbers.

Definition 2.5. A number $m \in \mathbb{Z}$ is *even* if there exists $k \in \mathbb{Z}$ such that

$$m = 2k.$$

On the other hand, m is *odd* if there exists a $k \in \mathbb{Z}$ such that

$$m = 2k + 1.$$

Definition 2.6 (Divisibility)**.** A number $a \in \mathbb{Z}$ is said to be *divisible* by a number $b \in \mathbb{Z}, b \neq 0$, if there exists a $k \in \mathbb{Z}$ such that

$$a = kb.$$

In this case we also say that b *divides* a, and write

$$b \mid a.$$

If b does *not* divide a, we write

$$b \nmid a.$$

Definition 2.7. For $n \in \mathbb{Z} \setminus \{0\}$, we denote the set of integers that are divisible by n by

$$n\mathbb{Z} = \{z \in \mathbb{Z} : z = nk, \quad k \in \mathbb{Z}\}.$$

2.2.2. *Strong induction.* In some proofs it is useful to assume all preceding cases rather than only the immediately preceding case. This gives an equivalent form of the principle of induction, called *strong induction.*

Theorem 2.8 (Strong induction)**.** *Let $n_0 \in \mathbb{N}$, and let $P(n)$ be a statement for each $n \in \mathbb{N}$ with $n \geq n_0$. Suppose that*

(1) $P(n_0)$ is true, and
(2) for every $n \geq n_0$, if $P(k)$ is true for every k with $n_0 \leq k \leq n$, then $P(n+1)$ is true.

Then $P(n)$ is true for every $n \geq n_0$.

Proof. Omitted. □

As an example where strong induction is natural, let us prove that all natural numbers $n \geq 2$ can be written as a product of *prime* numbers.

Theorem 2.9. *Every natural number $n \geq 2$ can be written as a product of prime numbers.*

Proof. When $n = 2$, we have $n \in \mathbb{P}$, so the result holds trivially. Now suppose $n \geq 2$ and the statement holds for all $2 \leq k \leq n$. If $n + 1 \in \mathbb{P}$, the result holds trivially, so suppose $n + 1 \notin \mathbb{P}$. Then we have

$$n + 1 = ab,$$

for some $2 \leq a, b \leq n$. By the induction hypothesis, a and b can be written as products of primes, so

$$a = p_1 p_2 \cdots p_i, \quad p_1, \ldots, p_i \in \mathbb{P},$$
$$b = q_1 q_2 \cdots q_j, \quad q_1, \ldots, q_j \in \mathbb{P}.$$

Thus we have

$$n + 1 = ab = p_1 p_2 \cdots p_i q_1 q_2 \cdots q_j.$$

The result follows from Theorem 2.8. □

2.2.3. *Sequences.*

Definition 2.10. A sequence of real numbers is a function $z \colon \mathbb{N} \to \mathbb{R}$, whose terms are denoted by $z_n = z(n)$. We write the sequence as

$$(z_n)_{n \in \mathbb{N}} = (z_0, z_1, z_2, \ldots).$$

Definition 2.11. A sequence of real numbers may also be specified recursively. First, for some $k \in \mathbb{Z}^+$, we define the initial terms $z_0, \ldots, z_{k-1}$. For each $n \geq k$, we then define z_n in terms of earlier terms $z_0, \ldots, z_{n-1}$.

2.3. **07.09.2026.** This lecture was mostly spent going through basic mathematical notation.

2.3.1. *Logical connectives.*

Definition 2.12. Let p and q be propositions. We define the following logical connectives:

(1) The *conjunction* $p \wedge q$, read "p and q", is true exactly when both p and q are true.
(2) The *disjunction* $p \vee q$, read "p or q", is true exactly when at least one of p and q is true.
(3) The *implication* $p \implies q$, read "p implies q", is false exactly when p is true and q is false.
(4) The *equivalence* $p \iff q$, read "p if and only if q", is true exactly when p and q have the same truth value.
(5) The *negation* $\neg p$, read "not p", is true exactly when p is false.

Definition 2.13. Let p and q be propositions.

(1) The *negated implication* $p \nRightarrow q$, read "p does not imply q", is true exactly when p is true and q is false.
(2) The *negated equivalence* $p \nLeftrightarrow q$, read "p is not equivalent to q", is true exactly when p and q have different truth values.

Definition 2.14. Let X be a set. A *predicate* $P(x)$ on X is a statement containing a variable x whose truth value depends on the chosen value of $x \in X$. For each $a \in X$, substituting a for x gives the proposition $P(a)$, which is either true or false. For example, $P(x) \coloneqq x > 0$ is a predicate on $\mathbb{R}$; $P(2)$ is true, whereas $P(-1)$ is false.

Definition 2.15. Let $P(x)$ be a predicate on a set X.

(1) The *universal quantification* $\forall x \in X \colon P(x)$, read "for every $x \in X$, $P(x)$", is true exactly when $P(a)$ is true for every $a \in X$.
(2) The *existential quantification* $\exists x \in X \colon P(x)$, read "there exists an $x \in X$ such that $P(x)$", is true exactly when $P(a)$ is true for at least one $a \in X$.

Definition 2.16. The *negated existential quantification* $\nexists x \in X \colon P(x)$, read "there does not exist an $x \in X$ such that $P(x)$", is true exactly when $P(a)$ is false for every $a \in X$. Equivalently,

$$\nexists x \in X \colon P(x) \quad \iff \quad \forall x \in X \colon \neg P(x).$$

2.3.2. *Basic set theory.*

Definition 2.17. A *set* is a collection of distinct objects, called its *elements*. If an object x is an element of a set A, we write $x \in A$; otherwise, we write $x \notin A$. The order in which the elements are listed, and any repetition among them, do not affect the set.

Definition 2.18. Two sets A and B are *equal*, written $A = B$, if they have exactly the same elements. Equivalently,

$$A = B \iff \forall x \colon (x \in A \iff x \in B).$$

Definition 2.19. The *empty set* is the set containing no elements. It is denoted by $\emptyset$.

Definition 2.20. Let A and B be are sets. A and B are said to be *disjoint*, if $A \cap B = \emptyset$.

Definition 2.21. A *singleton* set is a set consisting of exactly **one** element.

Definition 2.22. Let $P(x)$ be a predicate on a set X. The *set-builder notation*

$$A \coloneqq \{x \in X \colon P(x)\}$$

defines A to be the set of precisely those elements $x \in X$ for which $P(x)$ is true. Equivalently, for every $a \in X$,

$$a \in A \iff P(a).$$

Definition 2.23. A set A is a *subset* of a set B if every element of A is also an element of B. We then write $A \subseteq B$. If A is not a subset of B, we write $A \nsubseteq B$; this means that some element of A is not an element of B. If $A \subseteq B$ and $A \neq B$, then A is a *proper subset* of B, written $A \subset B$. Two sets are equal exactly when both containments hold:

$$A = B \iff A \subseteq B \wedge B \subseteq A.$$

Definition 2.24. Let A and B be subsets of a set X.

(1) The *union* of A and B is

$$A \cup B \coloneqq \{x \in X : x \in A \vee x \in B\}.$$

(2) The *intersection* of A and B is

$$A \cap B \coloneqq \{x \in X : x \in A \wedge x \in B\}.$$

(3) The *difference* of A and B is

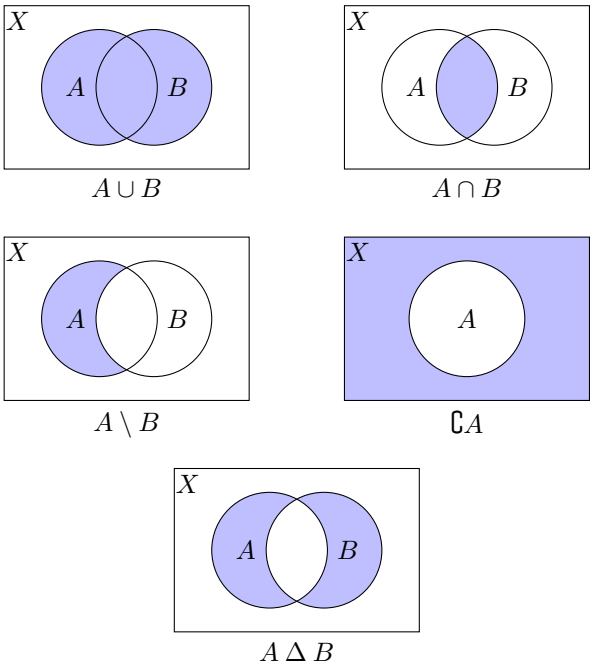
$$A \setminus B \coloneqq \{x \in X : x \in A \wedge x \notin B\}.$$

(4) The *complement* of A in X is

$$\complement A \coloneqq X \setminus A = \{x \in X : x \notin A\}.$$

(5) The *symmetric difference* of A and B is

$$A \,\Delta\, B \coloneqq (A \setminus B) \cup (B \setminus A).$$



2.4. **10.09.2026.**

2.4.1. *Power sets.*

Definition 2.25. The *power set* of a set A, denoted by $\mathcal{P}(A)$, is the set of all subsets of A:

$$\mathcal{P}(A) \coloneqq \{B : B \subseteq A\}.$$

In particular, $\emptyset \in \mathcal{P}(A)$ and $A \in \mathcal{P}(A)$.

Definition 2.26. Let A and B be sets. Their *Cartesian product* is the set of all ordered pairs whose first component belongs to A and whose second component belongs to B:

$$A \times B \coloneqq \{(a,b) : a \in A \wedge b \in B\}.$$

More generally, for sets $A_1, \dots, A_n$, their Cartesian product is

$$A_1 \times \cdots \times A_n \coloneqq \{(a_1, \dots, a_n) : a_i \in A_i \text{ for every } i \in \{1, \dots, n\}\}.$$

If all the factors are the same set A, one commonly writes

$$A^n \coloneqq \underbrace{A \times \cdots \times A}_{n \text{ factors}}.$$

Definition 2.27. Let I be a countably infinite *index set*. A family of subsets of a set X, indexed by I, is denoted by $(A_i)_{i \in I}$. Its *union* and *intersection* are defined by

$$\bigcup_{i \in I} A_i \coloneqq \{x \in X \colon \exists i \in I,\ x \in A_i\},$$
$$\bigcap_{i \in I} A_i \coloneqq \{x \in X \colon \forall i \in I,\ x \in A_i\}.$$

Thus, an element belongs to the union if it belongs to at least one of the sets, and it belongs to the intersection if it belongs to every one of them.

If the family itself is denoted by $\mathcal{A} = \{A_i : i \in I\}$, one also writes $\bigcup \mathcal{A}$ and $\bigcap \mathcal{A}$.

A common index set is $I = \mathbb{N}$. Since $0 \in \mathbb{N}$ by our convention, one writes

$$\bigcup_{n=0}^{\infty} A_n = \bigcup_{n \in \mathbb{N}} A_n, \qquad \bigcap_{n=0}^{\infty} A_n = \bigcap_{n \in \mathbb{N}} A_n.$$

For a family indexed by the positive integers, the corresponding notation is $\bigcup_{n=1}^{\infty} A_n$ and $\bigcap_{n=1}^{\infty} A_n$. When the index range is clear from context, these are also abbreviated as $\bigcup_n A_n$ and $\bigcap_n A_n$.

2.4.2. *Laws of set algebra.* Let A, B, and C be subsets of a universal set X. The union and intersection operations are *commutative*:

$$A \cup B = B \cup A, \qquad A \cap B = B \cap A.$$

They are also *associative*:

$$(A \cup B) \cup C = A \cup (B \cup C), \qquad (A \cap B) \cap C = A \cap (B \cap C).$$

The *distributive laws* are

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C),$$
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

The *De Morgan laws* relate complements to unions and intersections:

$$\complement(A \cup B) = \complement A \cap \complement B,$$
$$\complement(A \cap B) = \complement A \cup \complement B.$$

Finally, set difference can be expressed using intersection and complement:

$$A \setminus B = A \cap \complement B.$$

Equivalently, replacing B by C gives

$$A \setminus C = A \cap \complement C.$$

Theorem 2.28. *Let A and B be subsets of a universal set X. Then*

$$A \subseteq B \iff \complement B \subseteq \complement A.$$

Proof. Suppose first that $A \subseteq B$, and let $x \in \complement B$. Then $x \notin B$. Since every element of A belongs to B, it follows that $x \notin A$, and hence $x \in \complement A$. Therefore $\complement B \subseteq \complement A$.

Conversely, suppose that $\complement B \subseteq \complement A$, and let $x \in A$. If $x \notin B$, then $x \in \complement B$, so the assumed inclusion would imply that $x \in \complement A$, contradicting $x \in A$. Thus $x \in B$, and therefore $A \subseteq B$. □

2.5. **14.09.2026.**

2.5.1. *Proof by counterexample.* To disprove a universal claim, it is enough to find a single case in which the claim fails. Such a case is called a *counterexample*.

Example 2.29. Consider the claim that

$$(a+b)^2 = a^2 + b^2$$

for all real numbers a and b. Let $a = b = 1$. Then

$$(a+b)^2 = (1+1)^2 = 4, \qquad a^2 + b^2 = 1^2 + 1^2 = 2.$$

Since $4 \neq 2$, this is a counterexample, and therefore the claim is false.

2.5.2. *Proof by contradiction.* In a proof by contradiction, we assume that the claim is false and show that this assumption leads to a contradiction.

Example 2.30. Suppose that $A \subseteq B$ and $A \nsubseteq C$. We prove by contradiction that $B \nsubseteq C$.

Assume, to the contrary, that $B \subseteq C$. Since $A \nsubseteq C$, there exists an element $x \in A$ such that $x \notin C$. But $A \subseteq B$ implies that $x \in B$, and $B \subseteq C$ then implies that $x \in C$. This contradicts $x \notin C$. Therefore $B \nsubseteq C$.

2.5.3. *Relations.*

Definition 2.31. A *relation* from a set A to a set B is a subset

$$R \subseteq A \times B.$$

If $(a, b) \in R$, we say that a is related to b and write aRb. A relation on A is a relation from A to itself, that is, a subset of $A \times A$.

Definition 2.32. Let $R \subseteq A \times B$ be a relation. The *inverse relation* of R is

$$R^{-1} = \{(b, a) \in B \times A : (a, b) \in R\}.$$

Equivalently, $bR^{-1}a$ if and only if aRb.

Definition 2.33. Let R be a relation on a set A. The relation R is

(1) *reflexive* if xRx for every $x \in A$;
(2) *irreflexive* if $(x, x) \notin R$ for every $x \in A$;
(3) *symmetric* if, for all $x, y \in A$, xRy implies yRx;
(4) *antisymmetric* if, for all $x, y \in A$, xRy and yRx imply $x = y$;
(5) *asymmetric* if, for all $x, y \in A$, xRy implies $(y, x) \notin R$;
(6) *transitive* if, for all $x, y, z \in A$, xRy and yRz imply xRz; and
(7) *total* if, for all $x, y \in A$, either xRy or yRx.

A relation is an *equivalence relation* if it is reflexive, symmetric, and transitive.

2.6. **17.09.2026.**

2.6.1. *Properties of inverse relations.*

Theorem 2.34. *Let $R, S \subseteq A \times B$ be relations. Then*

(a) $(R^{-1})^{-1} = R$;
(b) $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$;
(c) $(R \cap S)^{-1} = R^{-1} \cap S^{-1}$;
(d) $(R \setminus S)^{-1} = R^{-1} \setminus S^{-1}$.

Proof. For (a), let $(a, b) \in A \times B$. By the definition of the inverse relation,

$$(a, b) \in (R^{-1})^{-1} \iff (b, a) \in R^{-1} \iff (a, b) \in R.$$

For the remaining identities, let $(b, a) \in B \times A$. Then

$$\begin{aligned}(b, a) \in (R \cup S)^{-1} &\iff (a, b) \in R \cup S \\ &\iff (a, b) \in R \vee (a, b) \in S \\ &\iff (b, a) \in R^{-1} \vee (b, a) \in S^{-1} \\ &\iff (b, a) \in R^{-1} \cup S^{-1},\end{aligned}$$

which proves (b). Similarly,

$$\begin{aligned}(b, a) \in (R \cap S)^{-1} &\iff (a, b) \in R \wedge (a, b) \in S \\ &\iff (b, a) \in R^{-1} \wedge (b, a) \in S^{-1} \\ &\iff (b, a) \in R^{-1} \cap S^{-1},\end{aligned}$$

proving (c). Finally,

$$\begin{aligned}(b, a) \in (R \setminus S)^{-1} &\iff (a, b) \in R \wedge (a, b) \notin S \\ &\iff (b, a) \in R^{-1} \wedge (b, a) \notin S^{-1} \\ &\iff (b, a) \in R^{-1} \setminus S^{-1},\end{aligned}$$

which proves (d). In each case, the two sets have exactly the same elements. □

2.6.2. *Composition of relations.*

Definition 2.35. Let $R \subseteq X \times Y$ and $S \subseteq Y \times Z$. Their *composite relation* is the relation from X to Z defined by

$$S \circ R \coloneqq \{(x, z) \in X \times Z : \exists y \in Y,\ (x, y) \in R \wedge (y, z) \in S\}.$$

Equivalently,

$$x(S \circ R)z \iff \exists y \in Y : xRy \wedge ySz.$$

Thus $S \circ R$ means that we first follow R, then S.

Theorem 2.36 (Associativity of composition)**.** *Let $R \subseteq X \times Y$, $S \subseteq Y \times Z$, and $T \subseteq Z \times U$. Then*

$$T \circ (S \circ R) = (T \circ S) \circ R.$$

Proof. Let $(x, u) \in X \times U$. If $(x, u) \in T \circ (S \circ R)$, there is a $z \in Z$ such that $x(S \circ R)z$ and zTu. The first relation supplies a $y \in Y$ with xRy and ySz. Since ySz and zTu, we have $y(T \circ S)u$, and hence $(x, u) \in (T \circ S) \circ R$.

Conversely, if $(x, u) \in (T \circ S) \circ R$, there is a $y \in Y$ such that xRy and $y(T \circ S)u$. Choose $z \in Z$ with ySz and zTu. Then $x(S \circ R)z$, so $(x, u) \in T \circ (S \circ R)$. Both inclusions hold, proving the equality. □

2.6.3. *Equivalence relations and equivalence classes.* Recall that a relation $\sim$ on a set X is an *equivalence relation* if it is reflexive, symmetric, and transitive.

Problem 2.37 (Lecture exercise)**.** Let $A = \{2, 3, 4, 5, 6\}$, and define a relation R on A by

$$xRy \iff x \text{ and } y \text{ have a common prime factor.}$$

Is R an equivalence relation?

Solution. The relation is reflexive: each $x \in A$ has a prime factor, and that prime divides both occurrences of x. More explicitly, 2 divides $2, 4, 6$, 3 divides 3, and 5 divides 5.

It is also symmetric. If xRy, some prime p divides both x and y. The same prime divides both y and x, so yRx.

However, it is not transitive. We have $4R6$, since 2 divides both 4 and 6, and $6R3$, since 3 divides both 6 and 3. But 4 and 3 have no common prime factor: their only prime factors are 2 and 3, respectively. Thus $(4, 3) \notin R$. Consequently, R is not an equivalence relation. □

Definition 2.38. Let $\sim$ be an equivalence relation on a set X, and let $a \in X$. The *equivalence class* of a is

$$[a]_\sim \coloneqq \{x \in X : x \sim a\}.$$

The element a is called a *representative* of this class. The set of all equivalence classes is the *quotient set*

$$X/\!\sim \coloneqq \{[a]_\sim : a \in X\}.$$

The following consequence explains why equivalence classes divide a set into separate parts; it supplements the definitions in the lecture.

Proposition 2.39. *Let $\sim$ be an equivalence relation on X. For every $a, b \in X$,*

(a) $a \in [a]_\sim$;
(b) $a \sim b$ if and only if $[a]_\sim = [b]_\sim$;
(c) the classes $[a]_\sim$ and $[b]_\sim$ are either equal or disjoint.

Therefore the equivalence classes form a partition *of X: they are nonempty, their union is X, and distinct classes are disjoint.*

Proof. Reflexivity gives $a \sim a$, proving (a).

For (b), suppose $a \sim b$. If $x \in [a]_\sim$, then $x \sim a$, so transitivity gives $x \sim b$. Hence $[a]_\sim \subseteq [b]_\sim$. Symmetry gives $b \sim a$, and the same argument gives the reverse inclusion. Conversely, if the classes are equal, then (a) gives $a \in [a]_\sim = [b]_\sim$, which means $a \sim b$.

For (c), if the classes intersect, choose an element c in their intersection. Then $c \sim a$ and $c \sim b$. By symmetry, $a \sim c$, so transitivity gives $a \sim b$. Part (b) now gives equality of the classes. Thus distinct classes must be disjoint. Finally, every class is a subset of X, and (a) shows that each element of X belongs to a nonempty class. Their union is therefore all of X, proving the partition claim. □

2.6.4. *Functions.*

Definition 2.40. Let X and Y be nonempty sets. A *function* (or *mapping*) f from X to Y assigns to each element $x \in X$ exactly one element $y \in Y$. We write

$$f \colon X \longrightarrow Y, \qquad y = f(x).$$

The set X is the *domain*, and Y is the *codomain*.

In terms of relations, the graph of f is

$$G_f = \{(x, f(x)) : x \in X\} \subseteq X \times Y.$$

A relation $F \subseteq X \times Y$ is the graph of a function precisely when, for every $x \in X$, there exists exactly one $y \in Y$ such that $(x, y) \in F$. Existence ensures that the function is defined at every input, and uniqueness ensures that its value is unambiguous.

2.7. **21.09.2026.**

2.7.1. *Inverse functions, images, and preimages.* The inverse relation of a function $f\colon X \to Y$ is a relation from Y to X, but it is not generally a function. If this inverse relation is a function from Y to X, it is called the *inverse function* of f, denoted by f^{-1}.

Definition 2.41. Let $f\colon X \to Y$ be a function. The *image* of a subset $A \subseteq X$ is

$$f(A) \coloneqq \{f(x) : x \in A\}.$$

The *preimage* of a subset $B \subseteq Y$ is

$$f^{-1}(B) \coloneqq \{x \in X : f(x) \in B\}.$$

Remark 2.42. The preimage $f^{-1}(B)$ is defined for every function f, even when f has no inverse function.

2.7.2. *Images and preimages of unions and intersections.*

Theorem 2.43. *Let $f\colon X \to Y$. For all $A, B \subseteq X$,*

(i) $f(A \cup B) = f(A) \cup f(B)$;
(ii) $f(A \cap B) \subseteq f(A) \cap f(B)$.

Theorem 2.44. *Let $f\colon X \to Y$. For all $A, B \subseteq Y$,*

(i) $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$;
(ii) $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$.

Proof. Let $A, B \subseteq Y$ and $x \in X$. For (i),

$$\begin{aligned} x \in f^{-1}(A \cup B) &\iff f(x) \in A \vee f(x) \in B \\ &\iff x \in f^{-1}(A) \vee x \in f^{-1}(B) \\ &\iff x \in f^{-1}(A) \cup f^{-1}(B). \end{aligned}$$

Similarly, for (ii),

$$\begin{aligned} x \in f^{-1}(A \cap B) &\iff f(x) \in A \wedge f(x) \in B \\ &\iff x \in f^{-1}(A) \wedge x \in f^{-1}(B) \\ &\iff x \in f^{-1}(A) \cap f^{-1}(B). \end{aligned}$$

Thus both identities hold. □

2.7.3. *Injective, surjective, and bijective functions.*

Definition 2.45. A function $f\colon X \to Y$ is *injective* if distinct inputs have distinct images: for all $x_1, x_2 \in X$,

$$x_1 \neq x_2 \implies f(x_1) \neq f(x_2).$$

Definition 2.46. A function $f\colon X \to Y$ is *surjective* if, for every $y \in Y$, there exists $x \in X$ such that $f(x) = y$. Equivalently, $f(X) = Y$.

Definition 2.47. A function $f\colon X \to Y$ is *bijective* if it is both injective and surjective.

Theorem 2.48. *A function $f\colon X \to Y$ has an inverse function $f^{-1}\colon Y \to X$ if and only if f is bijective.*

2.7.4. *Lecture exercise: the floor function.*

Problem 2.49 (Lecture exercise)**.** Let

$$f(x) = \lfloor x \rfloor = \max\{n \in \mathbb{Z} : n \le x\}.$$

Determine whether f is injective, surjective, or bijective for each of the following choices of domain and codomain:

(a) $f\colon \mathbb{R} \to \mathbb{Z}$;
(b) $f\colon \mathbb{Z} \to \mathbb{R}$;
(c) $f\colon \mathbb{Z} \to \mathbb{Z}$;
(d) $f\colon \mathbb{R} \to \mathbb{R}$.

Solution. (a) The function is not injective, since $f(1) = f(1.5) = 1$, although $1 \neq 1.5$. It is surjective: for every $y \in \mathbb{Z}$, any $x \in [y, y+1)$ satisfies $f(x) = y$. Therefore it is not bijective.

(b) For every $x \in \mathbb{Z}$, we have $f(x) = x$, so the function is injective. It is not surjective, since $1/2 \in \mathbb{R}$ is not the image of any integer. Therefore it is not bijective.

(c) Here $f(x) = x$ for all $x \in \mathbb{Z}$, so f is the identity function on $\mathbb{Z}$. It is both injective and surjective, and hence bijective.

(d) The function is neither injective nor surjective: $f(1) = f(1.5)$, and $1/2$ is not in its image. Consequently, it is not bijective. □

2.8. **24.09.2026.**

2.8.1. *Images followed by preimages.*

Theorem 2.50. *Let $f\colon X \to Y$, $A \subseteq X$, and $B \subseteq Y$. Then*

(i) $A \subseteq f^{-1}(f(A))$;
(ii) $f(f^{-1}(B)) \subseteq B$.

Proof. For (i), let $x \in A$. Then $f(x) \in f(A)$, so the definition of preimage gives $x \in f^{-1}(f(A))$.

For (ii), let $y \in f(f^{-1}(B))$. There exists $x \in f^{-1}(B)$ with $y = f(x)$. Since $x \in f^{-1}(B)$ means that $f(x) \in B$, we obtain $y \in B$. □

Remark 2.51. These inclusions need not be equalities. For $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, and $A = \{1\}$, we have $f^{-1}(f(A)) = \{-1, 1\}$. For the same function and $B = \{-1\}$, we have $f(f^{-1}(B)) = \emptyset$. Here f^{-1} denotes a preimage, not an inverse function.

2.8.2. *Composition of functions.*

Definition 2.52. Let $f\colon X \to Y$ and $g\colon Y \to Z$ be functions. Their *composite function* is

$$g \circ f\colon X \to Z, \qquad (g \circ f)(x) = g(f(x)) \quad (x \in X).$$

Thus we apply f first and g second.

Theorem 2.53. *If $f\colon X \to Y$ and $g\colon Y \to Z$ are bijections, then $g \circ f\colon X \to Z$ is a bijection.*

Proof. To prove injectivity, suppose that $(g \circ f)(x_1) = (g \circ f)(x_2)$. Then $g(f(x_1)) = g(f(x_2))$. Injectivity of g gives $f(x_1) = f(x_2)$, and injectivity of f gives $x_1 = x_2$.

To prove surjectivity, let $z \in Z$. Since g is surjective, there is $y \in Y$ with $g(y) = z$. Since f is surjective, there is $x \in X$ with $f(x) = y$. Thus $(g \circ f)(x) = g(y) = z$. □

2.8.3. *Identity maps and inverse functions.*

Definition 2.54. The *identity map* on a set X is the function

$$\mathrm{id}_X \colon X \to X, \qquad \mathrm{id}_X(x) = x \quad (x \in X).$$

Theorem 2.55. *Let $f \colon X \to Y$ and $g \colon Y \to X$. Then g is the inverse function of f if and only if*

$$g \circ f = \mathrm{id}_X \qquad \textit{and} \qquad f \circ g = \mathrm{id}_Y .$$

Proof. Suppose first that $g = f^{-1}$. For every $x \in X$, the inverse sends $f(x)$ back to x, so $g(f(x)) = x$. For every $y \in Y$, the value $g(y)$ is the unique element mapped to y by f, so $f(g(y)) = y$. These are the required identities.

Conversely, suppose both identities hold. If $f(x_1) = f(x_2)$, applying g gives

$$x_1 = g(f(x_1)) = g(f(x_2)) = x_2,$$

so f is injective. For every $y \in Y$, the element $g(y) \in X$ satisfies $f(g(y)) = y$, so f is surjective. Thus each y has exactly one preimage, namely $g(y)$, proving that $g = f^{-1}$. □

Corollary 2.56. *The inverse of a bijection is a bijection.*

Proof. Let $f \colon X \to Y$ be a bijection. If $f^{-1}(y_1) = f^{-1}(y_2)$, applying f gives $y_1 = y_2$, so f^{-1} is injective. For any $x \in X$, set $y = f(x)$. Then $f^{-1}(y) = x$, so f^{-1} is also surjective. □

2.8.4. *Infinite sets and their cardinalities.*

Definition 2.57. Sets A and B have *equal cardinality* (or are *equinumerous*) if there exists a bijection $A \to B$. We write $A \sim B$.

Definition 2.58. A set A is *countably infinite* if $A \sim \mathbb{N}$. Equivalently, its elements can be listed as $a_0, a_1, a_2, \ldots$, with every element appearing exactly once.

Remark 2.59. The Finnish term *numeroituva* in this lecture is defined by $A \sim \mathbb{N}$. We use "countably infinite" for this precise meaning; "countable" is also commonly used to include finite sets. Recall that $\mathbb{N} = \{0, 1, 2, \ldots\}$ in these notes.

For this discussion, we also allow functions with empty domain: the empty relation is the unique function $\emptyset \to B$ for any set B. In particular, $\mathrm{id}_\emptyset$ is a bijection from $\emptyset$ to itself. This extends the earlier convention of discussing functions between nonempty sets and allows us to include all subsets below.

Theorem 2.60. *Let X be a set. Equal cardinality is an equivalence relation on $\mathcal{P}(X)$.*

Proof. We verify reflexivity, symmetry, and transitivity.

Reflexivity. Let $A \subseteq X$. The identity map $\mathrm{id}_A \colon A \to A$ is injective, since equal outputs are equal inputs, and surjective, since each $a \in A$ is its own image. Hence $A \sim A$, including when $A = \emptyset$.

Symmetry. Let $A, B \subseteq X$ and suppose $A \sim B$. Choose a bijection $f \colon A \to B$. By corollary 2.56, its inverse $f^{-1} \colon B \to A$ is a bijection, so $B \sim A$.

Transitivity. Let $A, B, C \subseteq X$, and suppose $A \sim B$ and $B \sim C$. Choose bijections $f \colon A \to B$ and $g \colon B \to C$. By theorem 2.53, $g \circ f \colon A \to C$ is a bijection. Therefore $A \sim C$.

Thus $\sim$ is an equivalence relation on $\mathcal{P}(X)$. □

2.9. **28.09.2026.**

2.9.1. *Finite and countable sets.* We continue the discussion of cardinality. Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$. For $n \in \mathbb{N}$, write $[n] = \{1, \ldots, n\}$, with $[0] = \emptyset$.

Definition 2.61. A set A is *finite* if there is a bijection $[n] \to A$ for some $n \in \mathbb{N}$. Otherwise it is *infinite*. A set is *countable* if it is finite or countably infinite, and *uncountable* if it is not countable.

Remark 2.62. In today's lecture, the Finnish term *numeroituva* includes finite sets. This is the meaning of "countable" used here; "countably infinite" continues to mean equinumerous with $\mathbb{N}$.

Theorem 2.63. *The rational numbers are countably infinite:* $\mathbb{Q} \sim \mathbb{N}$.

Proof. First list the positive rational numbers in the array

$$\begin{array}{cccccc} 1/1 & 2/1 & 3/1 & 4/1 & \cdots \\ 1/2 & 2/2 & 3/2 & 4/2 & \cdots \\ 1/3 & 2/3 & 3/3 & 4/3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{array}$$

Visit the entries p/q, where $p, q \geq 1$, in increasing order of $p + q$, and within each diagonal in increasing order of p. Each diagonal has finitely many entries, so each pair (p, q) is visited after finitely many steps. Keep an entry only if its rational value has not already appeared. Every positive rational has a representation p/q, so it appears in the resulting list, and no value appears twice. The list is infinite because it contains every positive integer. Denote it by $q_1, q_2, q_3, \ldots$.

Define $f \colon \mathbb{N} \to \mathbb{Q}$ by

$$f(0) = 0, \qquad f(2n-1) = q_n, \qquad f(2n) = -q_n \quad (n \geq 1).$$

Every rational number is zero, positive, or negative, so f is surjective. The positive values are distinct, as are the negative values, and values from different sign classes cannot be equal. Thus f is injective as well, proving the claim. □

2.9.2. *Uncountability of the real numbers.*

Theorem 2.64 (Cantor's diagonal argument)**.** *The interval* $[0, 1]$*, and consequently* $\mathbb{R}$*, is uncountable.*

Proof. Suppose first that $[0, 1]$ is countable. It is infinite, since it contains the distinct numbers $1, 1/2, 1/3, \ldots$. Hence we could list all its elements as $x_1, x_2, x_3, \ldots$.

For each $x_n < 1$, choose its decimal expansion that is not eventually all 9's; use trailing zeros for terminating decimals. Represent 1 by $1.000\ldots$. Write

$$x_n = a_n.d_{n1}d_{n2}d_{n3}\ldots, \qquad d_{nj} \in \{0, 1, \ldots, 9\},$$

where a_n is the integer part. Define

$$b_n = \begin{cases} 2, & d_{nn} = 1, \\ 1, & d_{nn} \neq 1, \end{cases} \qquad y = \sum_{n=1}^{\infty} b_n 10^{-n} = 0.b_1b_2b_3\ldots.$$

Since $1 \leq b_n \leq 2$, the geometric series gives $1/9 \leq y \leq 2/9$, so $y \in [0, 1]$.

We check that the decimal comparison is valid even for numbers with two decimal representations. Fix n. If $x_n = 1$, then $x_n \neq y$. Otherwise, the decimal digits of x_n and y differ at position n, so they have a first differing position $k \leq n$. The difference at position k has absolute value at least 10^{-k}, while the difference of the remaining tails has absolute value at most

$$\sum_{j>k} 9 \cdot 10^{-j} = 10^{-k}.$$

Equality in this tail bound would require every later pair of digits to differ by 9, which is impossible because every digit of y is 1 or 2. The first difference therefore cannot be cancelled by the tail. Thus $y \neq x_n$ for every n, contrary to the assumption that the list contains all of $[0,1]$.

Finally, suppose $\mathbb{R}$ were countable. It is infinite, so it would have a listing without repetitions. Retain just the entries in $[0,1]$. Infinitely many entries remain, and every element of $[0,1]$ occurs at a finite position in the original list, hence also in the retained list. This would enumerate $[0,1]$, a contradiction. Therefore $\mathbb{R}$ is uncountable. □

2.9.3. *The size of a finite set.*

Lemma 2.65 (Pigeonhole principle)**.** *If $n, k \in \mathbb{N}$ and $k < n$, there is no injection $[n] \to [k]$.*

Proof. We use induction on k, proving the assertion for every $n > k$. For $k = 0$, no function from the nonempty set $[n]$ to the empty set exists.

Suppose the assertion holds for k, and let $n > k+1$. Assume there is an injection $f\colon [n] \to [k+1]$. Set $a = f(n)$, and let $\tau\colon [k+1] \to [k+1]$ interchange a and $k+1$, fixing all other elements. If $a = k+1$, take τ to be the identity. In either case $g = \tau \circ f$ is injective and $g(n) = k+1$. Injectivity implies $g(i) \neq k+1$ for every $i \in [n-1]$. Thus the restriction of g is an injection $[n-1] \to [k]$. Since $n - 1 > k$, this contradicts the induction hypothesis. The result follows by induction. □

Theorem 2.66. *For every finite set A, there is a unique $n \in \mathbb{N}$ such that $[n] \sim A$.*

Proof. Existence is the definition of finiteness. Suppose there are bijections $f\colon [n] \to A$ and $g\colon [k] \to A$, with $n \neq k$. By interchanging the names if necessary, assume $k < n$. Then $g^{-1} \circ f\colon [n] \to [k]$ is a bijection, hence an injection, contradicting lemma 2.65. Therefore $n = k$. □

Definition 2.67. The unique integer above is the *cardinality* (or *size*) of the finite set A, denoted by $|A|$ or $N(A)$. In particular, $|\emptyset| = 0$.

The following elementary fact ensures that the intersections and differences in our counting formulas are finite.

Lemma 2.68. *Every subset S of a finite set A is finite, and $|S| \leq |A|$.*

Proof. Choose a bijection $f\colon [n] \to A$. Scan the indices $1, \ldots, n$ in order, retaining precisely those for which $f(i) \in S$. This finite procedure produces a list $i_1 < \cdots < i_r$, where $0 \leq r \leq n$. The map $j \mapsto f(i_j)$ is a bijection $[r] \to S$: injectivity follows from that of f, and surjectivity follows because every element of S has an index that was retained. This also covers $S = \emptyset$, using the empty list. Hence $|S| = r \leq n$. □

2.9.4. *Unions and the inclusion–exclusion principle.*

Lemma 2.69 (Disjoint unions)**.** *If A and B are finite and disjoint, then $A \cup B$ is finite and $|A \cup B| = |A| + |B|$.*

Proof. Let $m = |A|$, $n = |B|$, and choose bijections $f\colon [m] \to A$ and $g\colon [n] \to B$. Define

$$h\colon [m+n] \to A \cup B, \qquad h(i) = \begin{cases} f(i), & 1 \leq i \leq m, \\ g(i-m), & m < i \leq m+n. \end{cases}$$

Each branch is injective, and values from different branches lie in disjoint sets, so h is injective. Every element of $A \cup B$ is reached by one of the branches, so h is surjective. Thus h is a bijection and the cardinality is $m + n$. The same construction works when either set is empty. □

Theorem 2.70. *Let A, B be finite sets. Then $A \cup B$ is finite and*

(a) $|A \cup B| = |A| + |B| - |A \cap B|$;
(b) if $A \cap B = \emptyset$, *then* $|A \cup B| = |A| + |B|$;
(c) $|A \cup B| = |A| + |B \setminus A|$;
(d) if $A \subseteq B$, *then* $|B| = |A| + |B \setminus A|$.

Proof. By lemma 2.68, both $B \setminus A$ and $A \cap B$ are finite. The decompositions

$$A \cup B = A \cup (B \setminus A), \qquad B = (A \cap B) \cup (B \setminus A)$$

are disjoint unions. Applying lemma 2.69 proves finiteness of $A \cup B$, part (c), and the identity

$$|B| = |A \cap B| + |B \setminus A|.$$

Substitute $|B \setminus A| = |B| - |A \cap B|$ into (c) to obtain (a). Part (b) follows from (a) and $|\emptyset| = 0$. If $A \subseteq B$, then $A \cup B = B$, so (c) gives (d). □

Theorem 2.71 (Inclusion–exclusion for three sets)**.** *Let* A, B, C *be finite sets. Then*

$$\begin{aligned} |A \cup B \cup C| = {} & |A| + |B| + |C| \\ & - |A \cap B| - |A \cap C| - |B \cap C| \\ & + |A \cap B \cap C|. \end{aligned}$$

Proof. Apply theorem 2.70(a) to $A \cup B$ and C:

$$|A \cup B \cup C| = |A \cup B| + |C| - |(A \cup B) \cap C|.$$

Distributivity gives $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$, so another application of the same formula yields

$$|(A \cup B) \cap C| = |A \cap C| + |B \cap C| - |A \cap B \cap C|.$$

Substituting this and $|A \cup B| = |A| + |B| - |A \cap B|$ into the first equality gives the stated formula. □

2.9.5. *Cardinality of Cartesian products.*

Theorem 2.72. *If* A, B *are finite sets, then* $A \times B$ *is finite and*

$$|A \times B| = |A||B|.$$

Proof. If either factor is empty, the product is empty and both sides are zero. Otherwise, write $A = \{a_1, \ldots, a_m\}$ with distinct elements. Then

$$A \times B = \bigcup_{i=1}^{m} (\{a_i\} \times B).$$

These sets are pairwise disjoint: an ordered pair in both the ith and jth sets would have first coordinate both a_i and a_j, forcing $i = j$. For each i, the map $B \to \{a_i\} \times B$, $b \mapsto (a_i, b)$, is a bijection, with inverse $(a_i, b) \mapsto b$. Thus each such set is finite and has cardinality $|B|$.

Repeated use of lemma 2.69 shows that a union of finitely many pairwise disjoint finite sets is finite and its size is the sum of their sizes. Formally, the assertion for one set is immediate; adding the next set preserves disjointness from the preceding union, so the two-set lemma proves the induction step. Consequently,

$$|A \times B| = \sum_{i=1}^{m} |\{a_i\} \times B| = m|B| = |A||B|. \qquad \square$$

Theorem 2.73. *Let* $n \geq 1$, *and let* $A_1, \ldots, A_n$ *be finite sets. Their Cartesian product is finite and*

$$|A_1 \times \cdots \times A_n| = \prod_{i=1}^{n} |A_i|.$$

Proof. We use induction on n. For $n = 1$, the identification $(a_1) \mapsto a_1$ is a bijection, so the formula holds. Suppose it holds for n factors. The map

$$(a_1, \dots, a_n, a_{n+1}) \longmapsto ((a_1, \dots, a_n), a_{n+1})$$

is a bijection from $A_1 \times \cdots \times A_{n+1}$ to $(A_1 \times \cdots \times A_n) \times A_{n+1}$, since its inverse simply removes the inner parentheses. The first factor is finite by the induction hypothesis. Hence theorem 2.72 and the induction hypothesis give

$$|A_1 \times \cdots \times A_{n+1}| = |A_1 \times \cdots \times A_n||A_{n+1}| = \prod_{i=1}^{n+1} |A_i|.$$

This proves both finiteness and the formula by induction, including cases with an empty factor. □

2.10. **01.10.2026.**

2.10.1. *The pigeonhole principle.*

Theorem 2.74 (Pigeonhole principle)**.** *If at least $k+1$ objects are placed in k boxes, where $k \geq 1$, then at least one box contains at least two objects.*

Proof. If each box contained at most one object, the k boxes would contain at most k objects altogether, a contradiction. □

Theorem 2.75 (Generalized pigeonhole principle)**.** *If $n \in \mathbb{N}$ objects are placed in $k \geq 1$ boxes, then at least one box contains at least $\lceil n/k \rceil$ objects. Here $\lceil x \rceil$ denotes the smallest integer greater than or equal to x.*

Proof. Put $r = \lceil n/k \rceil$. If every box contained fewer than r objects, each would contain at most $r - 1$. The total number of objects would then satisfy

$$n \leq k(r-1) < k\frac{n}{k} = n,$$

where the strict inequality follows from $r - 1 < n/k$. This is a contradiction. □

Example 2.76. There are 524 participants. How many are guaranteed to share a birthday, or a birth month?

There are 366 possible birthdays, allowing for February 29. By the generalized pigeonhole principle, at least

$$\left\lceil \frac{524}{366} \right\rceil = 2$$

participants share the same birthday (day and month). There are 12 possible birth months, so at least

$$\left\lceil \frac{524}{12} \right\rceil = 44$$

participants were born in the same month.

2.10.2. *Complex numbers.* The idea is to introduce a solution i of $x^2 + 1 = 0$, so that $i^2 = -1$. We make this precise using ordered pairs of real numbers.

Definition 2.77. The set of *complex numbers* is

$$\mathbb{C} = \{(a, b) : a, b \in \mathbb{R}\},$$

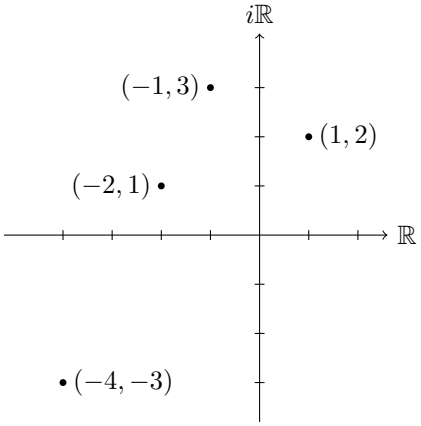
equipped with addition and multiplication defined by

$$\begin{aligned}(a, b) + (c, d) &= (a + c, b + d),\\ (a, b)(c, d) &= (ac - bd, ad + bc).\end{aligned}$$

Example 2.78. Consider the complex numbers $(1, 2)$ and $(-2, 1)$. Then

$$(1, 2) + (-2, 1) = (-1, 3),$$
$$(1, 2)(-2, 1) = (-2 - 2, 1 - 4) = (-4, -3).$$

They and their sum and product are shown in the complex plane below.



2.10.3. *Algebraic properties of complex numbers.*

Theorem 2.79. *The complex numbers form an abelian group under addition, with identity* $0 = (0, 0)$. *The nonzero complex numbers* $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$ *form an abelian group under multiplication, with identity* $1 = (1, 0)$. *Moreover,* $(\mathbb{C}, +, \cdot, 0, 1)$ *is a field.*

Proof. We check the group laws and distributivity directly from the pair definitions. Let $z = (a, b)$, $w = (c, d)$, and $u = (e, f)$.

Addition is closed in $\mathbb{C}$, and the corresponding laws in $\mathbb{R}$ give

$$z + w = (a + c, b + d) = (c + a, d + b) = w + z$$

and

$$(z + w) + u = (a + c + e, b + d + f) = z + (w + u).$$

Also, $z + (0, 0) = z$, and $z + (-a, -b) = (0, 0)$. Thus $(\mathbb{C}, +, 0)$ is an abelian group.

Multiplication is closed in $\mathbb{C}$ and commutative, since

$$zw = (ac - bd, ad + bc) = (ca - db, cb + da) = wz.$$

Expanding both parenthesizations gives

$$(zw)u = (ace - bde - adf - bcf,\ acf - bdf + ade + bce),$$
$$z(wu) = (ace - adf - bcf - bde,\ acf + ade + bce - bdf).$$

The coordinates agree, so multiplication is associative. Its identity is $(1, 0)$, because $(a, b)(1, 0) = (a, b)$. If $z \neq 0$, then $a^2 + b^2 > 0$, and

$$z^{-1} = \left(\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2}\right)$$

is nonzero and satisfies

$$zz^{-1} = \left(\frac{a^2 + b^2}{a^2 + b^2}, \frac{-ab + ba}{a^2 + b^2}\right) = (1, 0).$$

Products of nonzero elements are nonzero: if $zw = 0$ and $z \neq 0$, associativity gives $w = (z^{-1}z)w = z^{-1}(zw) = z^{-1}0 = 0$. Thus multiplication restricts to an abelian group on $\mathbb{C}^\times$.

Finally,

$$\begin{aligned} z(w+u) &= (a(c+e) - b(d+f),\ a(d+f) + b(c+e)) \\ &= (ac - bd, ad + bc) + (ae - bf, af + be) \\ &= zw + zu. \end{aligned}$$

Commutativity gives the other distributive law. Since $(0,0) \neq (1,0)$, these properties show that $\mathbb{C}$ is a field. □

2.10.4. *Real and imaginary parts.* We identify each $a \in \mathbb{R}$ with $(a,0) \in \mathbb{C}$. This identification preserves addition and multiplication, since

$$(a,0) + (c,0) = (a+c,0), \qquad (a,0)(c,0) = (ac,0).$$

Definition 2.80. For $z = (a,b) \in \mathbb{C}$, its *real part* and *imaginary part* are

$$\operatorname{Re} z = a, \qquad \operatorname{Im} z = b.$$

If $b = 0$, then z is real. If $a = 0$, then z is *purely imaginary* (including zero under this convention).

Definition 2.81. The complex number $i = (0,1)$ is the *imaginary unit*.

By the multiplication rule,

$$i^2 = (0,1)(0,1) = (-1,0) = -1.$$

Every complex number can therefore be written as

$$z = (a,b) = (a,0) + (0,b) = (a,0) + (b,0)(0,1) = a + bi.$$

Thus $\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}$.

Example 2.82. Let $z = a + bi$ and $w = c + di$. Then

$$\begin{aligned} z + w &= (a + bi) + (c + di) = (a + c) + (b + d)i, \\ zw &= (a + bi)(c + di) \\ &= ac + adi + bci + bdi^2 \\ &= (ac - bd) + (ad + bc)i. \end{aligned}$$

2.10.5. *Lecture exercise: complex arithmetic.*

Problem 2.83 (Lecture exercise)**.** Let $z = -3 + 2i$ and $w = 1 - 5i$. Calculate

(a) $z + w$;
(b) $z - w$;
(c) zw;
(d) z/w.

Solution. Using $i^2 = -1$, we obtain

$$\begin{aligned} z + w &= (-3 + 2i) + (1 - 5i) = -2 - 3i, \\ z - w &= (-3 + 2i) - (1 - 5i) = -4 + 7i, \\ zw &= (-3 + 2i)(1 - 5i) \\ &= -3 + 15i + 2i - 10i^2 = 7 + 17i. \end{aligned}$$

For the quotient, multiply numerator and denominator by $1+5i$:

$$\begin{aligned}\frac{z}{w} &= \frac{(-3+2i)(1+5i)}{(1-5i)(1+5i)} \\ &= \frac{-3-15i+2i+10i^2}{1-25i^2} \\ &= \frac{-13-13i}{26} = -\frac{1}{2}-\frac{1}{2}i.\end{aligned}$$

□

2.10.6. *Complex conjugation and modulus.*

Definition 2.84. The *complex conjugate* of $z = a+bi \in \mathbb{C}$ is

$$\overline{z} = a - bi.$$

Definition 2.85. The *modulus* of $z = a+bi \in \mathbb{C}$ is

$$|z| = \sqrt{a^2+b^2}.$$

It is the distance of (a,b) from the origin $(0,0)$ in the complex plane.

Proposition 2.86. *For every $z \in \mathbb{C}$,*

$$z\overline{z} = |z|^2, \qquad |z| = \sqrt{z\overline{z}}.$$

Proof. Write $z = a+bi$. Then

$$z\overline{z} = (a+bi)(a-bi) = a^2 - b^2i^2 = a^2+b^2 = |z|^2.$$

Taking the nonnegative real square root gives the second identity. □

2.11. **05.10.2026.**

2.11.1. *Properties of conjugation and modulus.*

Proposition 2.87. *For all $z, w \in \mathbb{C}$,*

(a) $\overline{z+w} = \overline{z} + \overline{w}$;
(b) $\overline{zw} = \overline{z}\,\overline{w}$;
(c) $\overline{\overline{z}} = z$;
(d) $\operatorname{Im} z = 0$ *if and only if* $\overline{z} = z$.

Proof. Write $z = a+bi$ and $w = c+di$, where $a,b,c,d \in \mathbb{R}$. Then

$$\begin{aligned}\overline{z+w} &= \overline{(a+c)+(b+d)i} = (a+c)-(b+d)i = \overline{z}+\overline{w}, \\ \overline{zw} &= \overline{(ac-bd)+(ad+bc)i} = (ac-bd)-(ad+bc)i \\ &= (a-bi)(c-di) = \overline{z}\,\overline{w}.\end{aligned}$$

Also, $\overline{\overline{z}} = \overline{a-bi} = a+bi = z$. Finally, $\overline{z} = z$ holds precisely when $a-bi = a+bi$, which is equivalent to $b = 0$, or $\operatorname{Im} z = 0$. □

Proposition 2.88. *For all $z, w \in \mathbb{C}$,*

(a) $|z| = \sqrt{z\overline{z}}$;
(b) $|zw| = |z||w|$;
(c) $zw = 0$ *if and only if* $z = 0$ *or* $w = 0$;
(d) $|z| = 0$ *if and only if* $z = 0$.

Proof. Part (a) was proved at the end of the previous lecture. Using conjugation of products and commutativity,

$$|zw|^2 = (zw)\overline{zw} = (z\overline{z})(w\overline{w}) = |z|^2|w|^2.$$

Moduli are nonnegative, so taking square roots proves (b). If $z = a+bi$, then $|z|^2 = a^2+b^2$. Since both squares are nonnegative, $|z| = 0$ exactly when $a = b = 0$, proving (d). Now (b) and (d) show that $zw = 0$ exactly when $|z||w| = 0$, which is equivalent to $z = 0$ or $w = 0$. This proves (c). □

2.11.2. *Reciprocals and quotients.*

Definition 2.89. If $z \in \mathbb{C} \setminus \{0\}$, its *reciprocal* is

$$z^{-1} = \frac{1}{z} = \frac{\overline{z}}{|z|^2}.$$

For $w \neq 0$, division is defined by

$$\frac{z}{w} = zw^{-1} = \frac{z\overline{w}}{|w|^2}.$$

Verification. Because $z \neq 0$, the preceding proposition gives $|z|^2 > 0$, so the formula is defined. Moreover,

$$z\frac{\overline{z}}{|z|^2} = \frac{z\overline{z}}{|z|^2} = 1.$$

Thus it agrees with the multiplicative inverse established earlier. The quotient formula follows by substituting the reciprocal of w. □

Example 2.90. If $z = a + bi \neq 0$, then

$$\frac{1}{a+bi} = \frac{a-bi}{a^2+b^2} = \frac{a}{a^2+b^2} - \frac{b}{a^2+b^2}i.$$

More generally, for $w = c + di \neq 0$,

$$\frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{c^2+d^2} = \frac{ac+bd}{c^2+d^2} + \frac{bc-ad}{c^2+d^2}i.$$

Proposition 2.91. *If $z \neq 0$, then $|z^{-1}| = |z|^{-1} = 1/|z|$.*

Proof. Since $zz^{-1} = 1$, multiplicativity of the modulus gives $|z||z^{-1}| = |1| = 1$. Divide by $|z| > 0$. □

2.11.3. *Polar form of a complex number.*

Definition 2.92. A *polar form* of $z \in \mathbb{C}$ is an expression

$$z = |z|(\cos\varphi + i\sin\varphi).$$

For $z \neq 0$, the angle φ is an *argument* of z, determined modulo 2π. If $z = a + bi$, then $\cos\varphi = a/|z|$ and $\sin\varphi = b/|z|$. Consequently, $\tan\varphi = b/a$ when $a \neq 0$, but this ratio alone does not determine the quadrant. For $z = 0$, every angle gives a polar form; its argument is left undefined.

Example 2.93. The numbers $z = 1+i$ and $w = \sqrt{3}-i$ have moduli $\sqrt{2}$ and 2, respectively. Their locations in the complex plane give arguments $\pi/4$ and $-\pi/6$. Thus

$$1 + i = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right), \qquad \sqrt{3} - i = 2\left(\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right).$$

Proposition 2.94. *If $z = |z|(\cos\varphi + i\sin\varphi)$, then*

$$\overline{z} = |z|(\cos(-\varphi) + i\sin(-\varphi)).$$

If, moreover, $z \neq 0$, then

$$z^{-1} = \frac{1}{|z|}(\cos(-\varphi) + i\sin(-\varphi)).$$

Proof. Conjugation changes the sign of the imaginary part, so

$$\overline{z} = |z|(\cos\varphi - i\sin\varphi) = |z|(\cos(-\varphi) + i\sin(-\varphi)),$$

using that cosine is even and sine is odd. For $z \neq 0$, the reciprocal formula gives

$$z^{-1} = \frac{\overline{z}}{|z|^2} = \frac{1}{|z|}(\cos(-\varphi) + i\sin(-\varphi)).$$

□

Proposition 2.95. *Let $z = |z|(\cos\alpha + i\sin\alpha)$ and $w = |w|(\cos\beta + i\sin\beta)$. Then*

$$zw = |z||w|(\cos(\alpha+\beta) + i\sin(\alpha+\beta)).$$

If $w \neq 0$, then

$$\frac{z}{w} = \frac{|z|}{|w|}(\cos(\alpha-\beta) + i\sin(\alpha-\beta)).$$

Proof. Expanding the product and using the angle addition identities,

$$\begin{aligned} zw &= |z||w|\big[(\cos\alpha\cos\beta - \sin\alpha\sin\beta) \\ &\qquad + i(\sin\alpha\cos\beta + \cos\alpha\sin\beta)\big] \\ &= |z||w|(\cos(\alpha+\beta) + i\sin(\alpha+\beta)). \end{aligned}$$

If $w \neq 0$, its reciprocal has modulus $1/|w|$ and angle $-\beta$ by the preceding proposition. Applying the product formula to zw^{-1} gives the quotient formula. □

2.12. **08.10.2026.**

2.12.1. *Integer powers and de Moivre's formula.*

Definition 2.96. For $z \in \mathbb{C}$ and a positive integer n, the power z^n is the product of n copies of z. For $z \neq 0$, define

$$z^0 = 1, \qquad z^{-n} = (z^n)^{-1} = \frac{1}{z^n}.$$

We also define $0^0 = 1$.

Theorem 2.97 (De Moivre's formula)**.** *If $z = r(\cos\theta + i\sin\theta) \neq 0$, where $r = |z|$, then for every $n \in \mathbb{Z}$,*

$$z^n = r^n\big(\cos(n\theta) + i\sin(n\theta)\big).$$

For $z = 0$, the formula also holds for positive integers n, with any choice of θ.

Proof. For $n = 0$, both sides are 1, and for $n = 1$ the formula is the polar form of z. If it holds for a positive integer n, the product formula for polar forms gives

$$z^{n+1} = z^n z = r^{n+1}\big(\cos((n+1)\theta) + i\sin((n+1)\theta)\big).$$

Induction proves the claim for all positive integers. If $n = -m$, where $m > 0$, the reciprocal formula gives

$$z^{-m} = \frac{1}{z^m} = r^{-m}\big(\cos(-m\theta) + i\sin(-m\theta)\big).$$

Finally, when $z = 0$ and $n > 0$, both sides vanish. □

Example 2.98 (Lecture exercise)**.** Compute $\left(\frac{1}{2} + \frac{1}{2}i\right)^{10}$.

Solution. The number $z = \frac{1}{2} + \frac{1}{2}i$ has modulus $1/\sqrt{2}$. Its real and imaginary parts are positive and equal, so an argument is $\pi/4$. Thus

$$z^{10} = \left(\frac{1}{\sqrt{2}}\right)^{10} \left(\cos\frac{10\pi}{4} + i\sin\frac{10\pi}{4}\right) = \frac{1}{32}\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right) = \frac{i}{32}. \qquad \square$$

2.12.2. *Euler's formula and exponential form.*

Definition 2.99. The *complex exponential* is defined by the power series

$$e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!}, \qquad z \in \mathbb{C}.$$

Its constant term is 1, including at $z = 0$.

Remark 2.100 (Analytic prerequisites). We take the power series for the real exponential, sine, and cosine as given; their derivation belongs to later analysis. The proof below uses these series and standard facts about absolutely convergent series. In particular, the exponential series converges absolutely for every z: for $z \neq 0$, the ratio of successive absolute terms is $|z|/(k+1) \to 0$, so the ratio test applies. At $z = 0$ convergence is immediate. Absolute convergence justifies grouping even and odd terms and taking Cauchy products below.

Theorem 2.101 (Euler's formula)**.** *For every $\theta \in \mathbb{R}$,*

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

Proof. Recall the real power series

$$\cos\theta = \sum_{j=0}^{\infty} \frac{(-1)^j\theta^{2j}}{(2j)!}, \qquad \sin\theta = \sum_{j=0}^{\infty} \frac{(-1)^j\theta^{2j+1}}{(2j+1)!}.$$

Since $i^{2j} = (-1)^j$ and $i^{2j+1} = i(-1)^j$, absolute convergence allows us to write

$$\begin{aligned} e^{i\theta} &= \sum_{k=0}^{\infty} \frac{(i\theta)^k}{k!} \\ &= \sum_{j=0}^{\infty} \frac{(-1)^j\theta^{2j}}{(2j)!} + i\sum_{j=0}^{\infty} \frac{(-1)^j\theta^{2j+1}}{(2j+1)!} \\ &= \cos\theta + i\sin\theta. \end{aligned}$$

$\square$

Proposition 2.102 (Exponential addition law)**.** *For all $u, v \in \mathbb{C}$,*

$$e^{u+v} = e^u e^v.$$

Consequently, for $x, y \in \mathbb{R}$,

$$e^{x+iy} = e^x(\cos y + i\sin y), \qquad |e^{x+iy}| = e^x > 0.$$

Proof. The Cauchy product of the absolutely convergent series gives

$$e^u e^v = \sum_{n=0}^{\infty}\sum_{k=0}^{n} \frac{u^k v^{n-k}}{k!(n-k)!} = \sum_{n=0}^{\infty} \frac{1}{n!}\sum_{k=0}^{n} \binom{n}{k} u^k v^{n-k} = \sum_{n=0}^{\infty} \frac{(u+v)^n}{n!} = e^{u+v}.$$

Here the inner sum is the binomial theorem. Taking $u = x$ and $v = iy$ and applying Euler's formula proves the second formula. Its modulus is $e^x\sqrt{\cos^2 y + \sin^2 y} = e^x$. $\square$

Definition 2.103. The *exponential form* of a nonzero complex number is

$$z = |z|e^{i\theta},$$

where θ is any argument of z. Arguments differ by integer multiples of 2π. We can write $0 = 0e^{i\theta}$ for any θ, but zero has no defined argument.

Remark 2.104. The addition law gives $e^0 = 1$, $e^{-u} = 1/e^u$, and $(e^u)^n = e^{nu}$ for all $n \in \mathbb{Z}$. Thus, for $z \neq 0$,

$$z^n = (|z|e^{i\theta})^n = |z|^n e^{in\theta} = |z|^n(\cos(n\theta) + i\sin(n\theta)),$$

giving a second proof of de Moivre's formula.

Example 2.105. Euler's identity and another exponential evaluation are

$$e^{i\pi} = -1, \qquad e^{i\pi} + 1 = 0, \qquad e^{-1+i\pi/2} = e^{-1}(0 + i) = \frac{i}{e}.$$

2.12.3. *Complex roots.*

Definition 2.106. Let n be a positive integer and $z \in \mathbb{C}$. An *nth root* of z is a number $w \in \mathbb{C}$ such that $w^n = z$.

Theorem 2.107. *Let $n \geq 1$ be an integer and let $z = re^{i\theta} \neq 0$, where $r = |z| > 0$. Then z has exactly n distinct nth roots, namely*

$$\begin{aligned} w_k &= r^{1/n} e^{i(\theta+2\pi k)/n} \\ &= r^{1/n}\left(\cos\frac{\theta + 2\pi k}{n} + i\sin\frac{\theta + 2\pi k}{n}\right), \qquad k = 0, \ldots, n-1. \end{aligned}$$

Here $r^{1/n}$ is the positive real nth root of r. The only nth root of zero is zero.

Proof. Each displayed number is a root, since de Moivre's formula yields

$$w_k^n = re^{i(\theta+2\pi k)} = re^{i\theta} = z.$$

Conversely, suppose $w^n = z \neq 0$. Write $w = se^{i\varphi}$ with $s > 0$. Taking moduli gives $s^n = r$, hence $s = r^{1/n}$. Equality of the unit complex factors gives $e^{in\varphi} = e^{i\theta}$, which is equivalent to $n\varphi - \theta \in 2\pi\mathbb{Z}$. Indeed, Euler's formula reduces this to $\cos(n\varphi - \theta) = 1$ and $\sin(n\varphi - \theta) = 0$. Thus $\varphi = (\theta + 2\pi k)/n$ for some $k \in \mathbb{Z}$. Replacing k by $k + n$ adds 2π to the angle and leaves w unchanged, so $k = 0, \ldots, n-1$ gives all roots.

If $w_k = w_\ell$, then $e^{2\pi i(k-\ell)/n} = 1$, so $k - \ell$ is divisible by n. For $0 \leq k, \ell < n$, this forces $k = \ell$, proving distinctness. Finally, $w^n = 0$ implies $|w|^n = 0$, hence $w = 0$. □

Remark 2.108. Changing the chosen argument θ by $2\pi m$, where $m \in \mathbb{Z}$, only permutes the root list. For $n \geq 3$, the roots are the vertices of a regular n-gon centred at the origin, on the circle of radius $r^{1/n}$; successive arguments differ by $2\pi/n$. For $n = 2$, the two roots are opposite points.

Example 2.109 (Lecture exercise, corrected)**.** Find all sixth roots of -8.

Solution. Here $r = |-8| = 8$, $\theta = \pi$, and $r^{1/6} = 8^{1/6} = \sqrt{2}$. Therefore

$$w_k = \sqrt{2}\, e^{i(\pi+2\pi k)/6}, \qquad k = 0, \ldots, 5.$$

Explicitly, these are

$$\sqrt{2}\, e^{i\pi/6}, \quad \sqrt{2}\, e^{i\pi/2}, \quad \sqrt{2}\, e^{5i\pi/6}, \quad \sqrt{2}\, e^{7i\pi/6}, \quad \sqrt{2}\, e^{3i\pi/2}, \quad \sqrt{2}\, e^{11i\pi/6}.$$

For example, $w_0 = \sqrt{6}/2 + i\sqrt{2}/2$ and $w_1 = i\sqrt{2}$. Each root satisfies $w_k^6 = (\sqrt{2})^6 e^{i(\pi+2\pi k)} = -8$. □

2.12.4. *Additional applications.* The following worked applications supplement the lecture material.

Example 2.110 (Rotations and repeated transformations)**.** Multiplication by $e^{i\alpha}$ rotates the complex plane anticlockwise through angle α: a point $re^{i\theta}$ becomes $re^{i(\theta+\alpha)}$. More generally, multiplication by $\rho e^{i\alpha}$, with $\rho > 0$, also scales distances from the origin by ρ.

For instance, the iteration $z_{m+1} = (\frac{1}{2} + \frac{1}{2}i)z_m$ has the explicit solution

$$z_m = 2^{-m/2} e^{im\pi/4} z_0, \qquad m \geq 0.$$

Each step rotates by $\pi/4$ and shrinks distances by $1/\sqrt{2}$. After eight steps, $z_8 = z_0/16$; after ten, $z_{10} = iz_0/32$. Since $|z_m| = 2^{-m/2}|z_0| \to 0$, every starting point approaches the origin.

Example 2.111 (Deriving trigonometric identities)**.** De Moivre's formula turns the binomial expansion of a complex power into identities for multiple angles. Writing $c = \cos\theta$ and $s = \sin\theta$, we obtain

$$\cos(3\theta) + i\sin(3\theta) = (c + is)^3 = (c^3 - 3cs^2) + i(3c^2 s - s^3).$$

Equating real and imaginary parts and using $c^2 + s^2 = 1$ gives

$$\cos(3\theta) = 4\cos^3\theta - 3\cos\theta, \qquad \sin(3\theta) = 3\sin\theta - 4\sin^3\theta.$$

This method generalises to any positive integer multiple of an angle.

Example 2.112 (Roots of unity and cancellation)**.** For $n \geq 2$, set $\zeta = e^{2\pi i/n}$. The nth roots of unity are $1, \zeta, \ldots, \zeta^{n-1}$. Since $\zeta^n = 1$ and $\zeta \neq 1$, the finite geometric sum gives

$$1 + \zeta + \cdots + \zeta^{n-1} = \frac{1 - \zeta^n}{1 - \zeta} = 0.$$

Thus equally spaced unit vectors cancel. Taking real and imaginary parts gives

$$\sum_{k=0}^{n-1} \cos\frac{2\pi k}{n} = 0, \qquad \sum_{k=0}^{n-1} \sin\frac{2\pi k}{n} = 0.$$

More generally, every integer m satisfies

$$\sum_{k=0}^{n-1} \zeta^{mk} = \begin{cases} n, & n \mid m, \\ 0, & n \nmid m. \end{cases}$$

If $n \mid m$, every summand is 1; otherwise, apply the geometric sum to $\zeta^m \neq 1$. This cancellation is the basic algebra behind separating discrete frequencies in the discrete Fourier transform.

Example 2.113 (Solving and factoring a polynomial)**.** To solve $w^3 = -8$, use argument π and root modulus 2. The roots are

$$2e^{i\pi/3} = 1 + i\sqrt{3}, \qquad 2e^{i\pi} = -2, \qquad 2e^{5i\pi/3} = 1 - i\sqrt{3}.$$

Pairing the conjugate roots gives a real quadratic:

$$(X - (1 + i\sqrt{3}))(X - (1 - i\sqrt{3})) = (X - 1)^2 + 3 = X^2 - 2X + 4.$$

Hence

$$X^3 + 8 = (X + 2)(X^2 - 2X + 4) = (X + 2)(X - 1 - i\sqrt{3})(X - 1 + i\sqrt{3}).$$

Complex roots thus describe the complete linear factorisation even when only one root is real.

2.13. **12.10.2026.**

2.13.1. *Solving quadratic equations over the complex numbers.*

Proposition 2.114 (The quadratic formula)**.** *Let $a, b, c \in \mathbb{R}$, with $a \neq 0$. The solutions in $\mathbb{C}$ of*

$$ax^2 + bx + c = 0$$

are determined by the discriminant *$D = b^2 - 4ac$:*

- *If $D > 0$, there are two distinct real roots, $x = (-b \pm \sqrt{D})/(2a)$.*
- *If $D = 0$, there is one distinct root, $x = -b/(2a)$, called a* double root*.*
- *If $D < 0$, there are two nonreal complex roots,*

$$x_1 = \frac{-b + i\sqrt{-D}}{2a}, \qquad x_2 = \frac{-b - i\sqrt{-D}}{2a}.$$

These roots are complex conjugates: $x_1 = \overline{x_2}$.

Example 2.115. Solve $3x^2 - 5x + 2 = 0$. Here $D = 25 - 24 = 1$, so

$$x = \frac{5 \pm \sqrt{25 - 24}}{6} = \frac{5 \pm 1}{6}.$$

Thus the roots are $x = 1$ and $x = 2/3$.

Example 2.116. Solve $x^2 + x + 1 = 0$. Since $D = 1 - 4 = -3$, the roots are

$$x = \frac{-1 \pm i\sqrt{3}}{2},$$

or, explicitly,

$$x_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \qquad x_2 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i = \overline{x_1}.$$

2.13.2. *Biquadratic equations.* An equation of the form

$$az^4 + bz^2 + c = 0, \qquad a \neq 0,$$

can be reduced to a quadratic equation by the substitution $x = z^2$. First solve

$$ax^2 + bx + c = 0$$

for x_1, x_2, and then solve $z^2 = x_1$ and $z^2 = x_2$ over $\mathbb{C}$. Each nonzero value of x has two square roots; zero has only the square root zero. Repeated values are counted only once when listing distinct solutions.

2.13.3. *Lecture exercise: fourth roots.*

Problem 2.117 (Lecture exercise)**.** Solve in $\mathbb{C}$:

$$z^4 + 1 - i\sqrt{3} = 0.$$

Solution. Rewrite the equation as $z^4 = -1 + i\sqrt{3}$. Set $w = -1 + i\sqrt{3}$. Its modulus is

$$|w| = \sqrt{(-1)^2 + (\sqrt{3})^2} = 2.$$

Since w lies in the second quadrant, an argument is $2\pi/3$. Hence

$$w = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right) = 2e^{2\pi i/3}.$$

The four roots are therefore

$$\begin{aligned} z_k &= 2^{1/4}\left(\cos\frac{2\pi/3 + 2\pi k}{4} + i\sin\frac{2\pi/3 + 2\pi k}{4}\right) \\ &= 2^{1/4}e^{i(\pi/6 + k\pi/2)}, \qquad k = 0, 1, 2, 3. \end{aligned}$$

In Cartesian form,

$$z_0 = 2^{1/4}\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right), \qquad z_1 = 2^{1/4}\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right),$$
$$z_2 = 2^{1/4}\left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i\right), \qquad z_3 = 2^{1/4}\left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right).$$

Indeed, $z_k^4 = 2e^{i(2\pi/3+2\pi k)} = -1 + i\sqrt{3}$ for each k. These are four distinct roots, so the list is complete. □